Practice assignment 12 - Not Graded

This assignment will not be graded and is only for practice.
Note : This activity is only for practice purpose and it will not be counted towards the Final score.

1 point
Let $f(x, y)$f(x, y) be defined in a neighbourhood of (a, b)(a, b) and $D(x, y) = f_{xx} f_{yy} - f_{xy}^2$ D(x, y) = fxxfyy − fxy2. Suppose (a, b)(a, b) is a critical point of ff. Which of the following options is(are) true?

- [ ] If $D(a, b) > 0$D(a, b) > 0 and $f_{xx}(a, b) > 0$fxx(a, b) > 0, then (a, b)(a, b) is a point of local minimum.
- [ ] If $D(a, b) < 0$D(a, b) < 0, then (a, b)(a, b) is a point of local maximum.
- [ ] If $D(a, b) = 0$D(a, b) = 0, then (a, b)(a, b) may be a point of minimum or maximum or neither.
- [ ] If $D(a, b) < 0$D(a, b) < 0, then (a, b)(a, b) is a saddle point.

No, the answer is incorrect.
Score: 0
Accepted Answers:
If $D(a, b) > 0$D(a, b) > 0 and $f_{xx}(a, b) > 0$fxx(a, b) > 0, then (a, b)(a, b) is a point of local minimum.
If $D(a, b) = 0$D(a, b) = 0, then (a, b)(a, b) may be a point of minimum or maximum or neither.
If $D(a, b) < 0$D(a, b) < 0, then (a, b)(a, b) is a saddle point.

1 point
Choose the correct statements about $f(x, y) = x^4 + y^4$f(x, y) = x4 + y4, where $D(x, y) = f_{xx} f_{yy} - f_{xy}^2$ D(x, y) = fxxfyy − fxy2.

- [ ] $(0, 0)$(0, 0) is the only critical point.
- [ ] $D(0, 0) > 0$D(0, 0) > 0 and $f_{xx}(0, 0) < 0$fxx(0, 0) < 0 and so $(0, 0)$(0, 0) is a point of local maximum.
- [ ] $D(0, 0) = 0$D(0, 0) = 0.
- [ ] $(0, 0)$(0, 0) is a point of local minimum.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$(0, 0)$(0, 0) is the only critical point.
$D(0, 0) = 0$D(0, 0) = 0.
$(0, 0)$(0, 0) is a point of local minimum.

1 point
Let $f(x, y) = (9x^2 - 1)(1 + 4y)$f(x, y) = (9x2 − 1)(1 + 4y) be defined on the box $-2 \le x \le 3$−2 ≤ x ≤ 3 and $-1 \le y \le 4$−1 ≤ y ≤ 4. Then which of the following options is (are) true?

- ○ $(0, 1)$(0, 1) is a point of local minimum.
- ○ $(3, 4)$(3, 4) is a point of absolute maximum.

○
−105−105 is the absolute minimum.
○
$(-2, -1)$(−2,−1) is a point of absolute minimum.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$(3, 4)$(3, 4) is a point of absolute maximum.

1 point

Suppose $z = f(x) + yg(x)$z = f(x) + yg(x) for some polynomial functions ff and gg. Which of the following options is true?

○
z_{yy} = 0zyy = 0
○
z_{yy} zyy is a non-constant function of xx.
○
z_{yy} zyy is a non-constant function of yy.
○
z_{yy} = 1zyy = 1

No, the answer is incorrect.
Score: 0
Accepted Answers:
z_{yy} = 0zyy = 0

1 point

Define a function

$$f(x, y) = \begin{cases} \frac{xy(x^2 - y^2)}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$f(x, y) = {x2+y2xy(x2−y2) if(x, y) = (0, 0)0 if(x, y) = (0, 0)

Choose the set of correct options.

☐
$f_{xy}(0, 0) = -1$fxy(0, 0) = −1
☐
$f_{yx}(0, 0) = 1$fyx(0, 0) = 1
☐
$f_{xy}(0, 0) = f_{yx}(0, 0)$fxy(0, 0) = fyx(0, 0)
☐
$f_x(0, 0) = f_y(0, 0)$fx(0, 0) = fy(0, 0)

No, the answer is incorrect.
Score: 0
Accepted Answers:
$f_{xy}(0, 0) = -1$fxy(0, 0) = −1
$f_{yx}(0, 0) = 1$fyx(0, 0) = 1
$f_x(0, 0) = f_y(0, 0)$fx(0, 0) = fy(0, 0)

1 point

Consider a function $f(x, y, z) = x^3 y + yz$f(x, y, z) = x3y + yz. If AA is the the Hessian matrix of $f(x, y, z)$f(x, y, z) at (1,1,0), then which of the following options is(are) true?

☐
$det(A) = 0$det(A) = 0

☐ $Rank(A)$ = 2Rank(A) = 2

☐ $Nullity(A)$ = 1Nullity(A) = 1

☐ $det(A) = -6$det(A) = −6

☐ $Rank(A)$ = 3Rank(A) = 3

No, the answer is incorrect.
Score: 0
Accepted Answers:
$det(A) = -6$det(A) = −6
$Rank(A)$ = 3Rank(A) = 3

1 point

Let $f: R^6 \rightarrow R$f : R6 → R be a polynomial function. Which of the following options is true?

○ Hessian matrix of the function is a symmetric matrix of order 6.

○ All possible mixed partial derivatives of second order have the same value at a point of R^6 R6

○ Hessian matrix of the function is a symmetric matrix of order 5.

○ Hessian matrix of the function is a symmetric matrix of order 4.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Hessian matrix of the function is a symmetric matrix of order 6.

Let S be a rectangular box. Let x, yx, y and zz be the length of the sides of the box and volume of the box $V = xyz$V = xyz. If dd is the length of the diagonal of the rectangular box, then $d = \sqrt{x^2 + y^2 + z^2}$d = x2 + y2 + z2
√
.

Use this information to answer the questions 8 and 9.

1 point

If the diagonal is of length 2 units and the volume of the box is the maximum possible, then which of the following options is (are) true?

☐ SS is a cube.

☐ All sides are of different length.

☐ $x = \frac{2}{\sqrt{3}}$x = 3
√
2

☐ $y = \sqrt{\frac{1}{3}}$y = 31
√

No, the answer is incorrect.
Score: 0
Accepted Answers:

SS is a cube.
$x = \frac{2}{\sqrt{3}}$x = 3
√
2
1 point
If the length of the diagonal of the box is 3 units, then which of the followings is the maximum possible volume of the box?

○ √3 3 √ cube units.

○ 3√33 3 √ cube units.

○ 1 cube units.

○ 33 cube units

No, the answer is incorrect.
Score: 0
Accepted Answers:
3√33 3 √ cube units.

1 point
Consider a function $f(x, y)$f(x, y).Let v_1, v_2, v_3 v1, v2, v3 be the critical points of the function ff and H_k Hk be the Hessian matrix of the function at the point v_k vk.Then which of the following options is(are) true?

☐ If $H_1 = \begin{bmatrix} 1 & 0 \\ -3 & 2 \end{bmatrix}$H1 = [1−302], then v_1 v1 is a saddle point.

☐ If $H_2 = \begin{bmatrix} 1 & 1 \\ 3 & 2 \end{bmatrix}$H2 = [1312], then v_2 v2 is a saddle point.

☐ If $H_3 = \begin{bmatrix} 3 & 1 \\ -3 & 1 \end{bmatrix}$H3 = [3−311], then v_3 v3 is a point of local extremum.

☐ If $H_1 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$H1 = [2002], then v_1 v1 is a point of local minimum.

No, the answer is incorrect.
Score: 0
Accepted Answers:
If $H_2 = \begin{bmatrix} 1 & 1 \\ 3 & 2 \end{bmatrix}$H2 = [1312], then v_2 v2 is a saddle point.

If $H_3 = \begin{bmatrix} 3 & 1 \\ -3 & 1 \end{bmatrix}$H3 = [3−311], then v_3 v3 is a point of local extremum.

If $H_1 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$H1 = [2002], then v_1 v1 is a point of local minimum.

1 point
Consider the following function

$f(x, y, z) = x^2 y^2 + xz^2$ f(x, y, z) = x2y2 + xz2

Which of the following options is (are) true?

☐ $(0, y, 0),$ where $y \in R(0, y, 0),$ where y ∈ R are critical points.

☐ $(0, 0, z),$ where $z \in R(0, 0, z),$ where z ∈ R are critical points.

☐ $(1, y, 0)$ where $y \in R(1, y, 0)$ where y ∈ R are critical points.

☐ $(0, y, 1),$ where $y \in R(0, y, 1),$ where y ∈ R are critical points.

☐ $(x, 0, 0),$ where $x \in R(x, 0, 0),$ where x ∈ R are critical points.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(0, y, 0),$ where $y \in R(0, y, 0),$ where y ∈ R are critical points.

$(x, 0, 0),$ where $x \in R(x, 0, 0),$ where x ∈ R are critical points.

1 point

Consider a function $f(x, y, z)$f(x, y, z). Let v_1, v_2, v_3 v1, v2, v3 be the critical points of the function f f and H_k Hk be the Hessian matrix of the function at the point v_k vk.Then which of the following options is(are) true?

☐ If $H_1 = \begin{bmatrix} -1 & 0 & 0 \\ 1 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix}$H1 = $\begin{bmatrix} & & \\ -1 & 1 & 0 \\ 1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$, then v_1 v1 is a saddle point.

☐ If $H_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$H2 = $\begin{bmatrix} & & \\ 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, then v_2 v2 is a saddle point.

☐ If $H_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$H3 = $\begin{bmatrix} & & \\ 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, then v_3 v3 is a point of local maximum.

☐ If $H_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$H1 = $\begin{bmatrix} & & \\ 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, then v_1 v1 is a point of local minimum.

☐ If $H_2 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix}$H2 = $\begin{bmatrix} & & \\ -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix}$, then v_2 v2 is a point of local maximum.

☐

If $H_3 = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$H3 = $\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}$, then v_3 v3 is a point of local minimum.

No, the answer is incorrect.
Score: 0
Accepted Answers:

If $H_1 = \begin{bmatrix} -1 & 0 & 0 \\ 1 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix}$H1 = $\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}$, then v_1 v1 is a saddle point.

If $H_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$H1 = $\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}$, then v_1 v1 is a point of local minimum.

If $H_2 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix}$H2 = $\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}$, then v_2 v2 is a point of local maximum.

Check Answers